TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an sLLM-Based Autonomous SecOps Agent with Unified Cyber Defense Platform**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	*Network System Security*	*Software*	*Software Solutions*
Concept of Proposed Technology	**Core Concept** The proposed project aims to develop an Autonomous Active Security Agent by integrating advanced analytics, AI-driven reasoning, and automated cyber-response technologies into a unified Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) framework. The core concept is to construct a next-generation security operations architecture in which high-fidelity data abstraction, threat detection models, and autonomous response mechanisms operate cohesively within a single execution environment. By leveraging proprietary big-data processing architecture, digital-forensic–based data interpretation methods, and generative AI–assisted operational intelligence, the project seeks to establish a cyber-defense framework capable of understanding diverse security data sources, identifying complex or emerging threats, and orchestrating response actions with minimal human intervention. The ultimate goal is to enable a self-evolving, AI-centered security agent that improves detection accuracy, accelerates response speed, and reduces operational workload across varied			

	IT environments. ○ **Key Functions and Potential Applications** 1. High-Performance Unified SIEM & SOAR Platform A proprietary storage, indexing, and query engine enables integrated event analysis and automated response within a unified architectural stack. This reduces inter-system delays and supports fast data correlation, threat reasoning, and immediate action. 2. Digital Forensic Core-Based Data Abstraction Advanced forensic interpretation modules convert unstructured system artifacts—file systems, registry elements, logs, and execution traces—into structured data streams suitable for machine reasoning. This foundation enables development of high-resolution threat-detection AI without additional pre-processing burdens. 3. Generative AI for Security Operations sLLM-based features such as natural-language query transformation, automated incident summarization, and context-aware reasoning provide a basis for developing more sophisticated detection logic and autonomous mitigation workflows. 4. Real-Time Multi-Source Data Integration Support for heterogeneous data types—including logs, network packets, endpoint telemetry, and cloud-native data—allows AI models to operate across diverse infrastructures. Built-in connectors enable rapid integration into various security environments for iterative development and evaluation. 5. Modular Application Ecosystem for Deployment and Expansion A dedicated app distribution framework allows newly developed detection models, automated playbooks, and data collectors to be packaged, deployed, and updated as modular components. This supports scalable adoption of the technologies developed through the project.
Rationale for Joint R&D Project	Developing an “Intelligent Autonomous Security Agent” involves overcoming several fundamental technical barriers that cannot be resolved through software engineering alone. 1. Combining Engineering Limits with Domain Expert Tacit Knowledge Detecting sophisticated threats cannot be solved by data processing speed alone. The intuition to read an attacker's subtle psychology, identify context missed by machines, and filter out false positives belongs to the realm of tacit knowledge held by experienced security analysts. To enable an AI Agent to judge like a human beyond simple automation, the process of formalizing this field know-how and injecting it into the model must be prioritized through collaboration. 2. Overcoming Commercial Development Limits via Advanced Model Optimization Theory To deploy the powerful intelligence of large language models (sLLM) onto resource-constrained on-premise security appliances, we must solve the difficult challenge of drastically reducing model size while maintaining inference performance. While corporate research labs must focus on immediate product stability, advanced technologies such as Knowledge Distillation, Quantization, and Sparsity Control require deep mathematical understanding and theoretical verification. Synergistic collaboration with academia is

	essential to "transfuse" rapidly evolving AI optimization theories and implement them at a commercial product level. 3. Resolving Defense Asymmetry Against AI-Driven Intelligent Attacks (Adversarial AI) Attackers are already actively exploiting Generative AI to create infinite variants of malware and utilize Adversarial AI techniques to bypass defense systems. It is nearly impossible for an individual security company to catch up with this speed of technological evolution alone. Through international joint research, we must accelerate the R&D speed of defense technologies and build a federated learning system that shares and learns multi-national Cyber Threat Intelligence (CTI) in real time. This is critical to closing the AI Capability Gap between attackers and defenders at an early stage. ○ **Why Global Collaboration?** These barriers require world-class academic expertise in cybersecurity, AI optimization, adversarial machine learning, and autonomous decision-making—fields in which Purdue University is internationally recognized as a global leader. Purdue's excellence is represented by institutions such as the Center for Education and Research in Information Assurance and Security (CERIAS), one of the world's premier cybersecurity research centers; the Systems and Software Security Lab (PurS³), which advances system and application security; and the Applied Machine Intelligence Lab, which specializes in AI reasoning, model optimization, and autonomous intelligent systems. This globally competitive research ecosystem provides scientific depth and advanced methodologies that cannot be replicated within a single company. Therefore, joint R&D with Purdue University is essential to overcome the identified barriers and to accelerate the development of a truly Intelligent Autonomous Security Agent capable of defending against next-generation cyber threats.
Key Words	1. sLLM 2. AI Threat Detection 3. Security Automation 4. SecOps 5. SIEM/SOAR

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

The objective of this project is to develop an Intelligent Autonomous Security Agent capable of understanding multi-source security data, detecting complex and emerging threats, and executing real-time, context-aware response actions with minimal human intervention. This will be achieved by integrating advanced AI reasoning models, optimized lightweight sLLMs, digital-forensic–based data abstraction, and automated security orchestration into a unified cyber-defense framework.

1. Development of Autonomous Cyber Threat Hunting & Response Agent

Going beyond traditional passive defense (detecting logs upon arrival), the goal is to implement an active security system where sLLM analyzes the latest vulnerability (CVE) information to generate valid exploit scenarios and simulates them against the defense system to self-validate the integrity of detection rules.

- Generative CVE Analysis & Adversarial Emulation
  - Technology where sLLM reads unstructured text from public CVE (Common Vulnerabilities and Exposures) documents and latest threat campaign reports to autonomously generate valid Attack Payloads and mock hacking scenarios capable of triggering the vulnerabilities.
  - Automated Exploit Generation: Generating executable attack codes in the form of Python scripts or packet dumps based on text-described vulnerability information.
  - Virtual Attack Simulation: Internalizing BAS (Breach and Attack Simulation) functions to project generated attacks into a virtual environment (Sandbox) or testbed to verify if the current security equipment can block them.
- Self-Validating Rule Generation & Optimization
  - Implementing a Closed-loop system that generates optimal detection rules based on simulated attack data, validates whether the generated rules block actual attacks, and then deploys them.
  - Rule-Testing Loop: AI iteratively performs the process of "Attack Generation -> Rule Generation → Simulation → Rule Modification upon Detection Failure → Re-validation" to secure Zero-Error Rules.
  - Shadow Rule Mode: Instead of applying generated rules immediately in blocking mode, they are first applied virtually (Shadowing) to real traffic to monitor for false positives before activation.
- Autonomous Forensic Reconstruction & Response
  - Reconstructing the full scope of an attack into a visual timeline by connecting fragmented logs and artifacts upon actual attack occurrence, and autonomously responding in integration with SOAR.
  - Context-Aware Investigation: Instead of viewing a single alert event in isolation, sLLM analyzes all related events before and after to generate an automated forensic report explaining "Who did what, how, and where they moved."

- Dynamic Playbook Execution: Dynamically combining and executing the most appropriate response actions (isolation, blocking, account lockout, etc.) according to the attack type and damage scope, rather than following fixed response procedures.

2. Autonomous Privacy Identification & Dynamic Observability for Massive Heterogeneous DBs

A technology to precisely identify hidden personal information (Shadow Data) using sLLM without affecting operational performance (Non-intrusive) and secure visibility by tracking schema changes in real-time within enterprise environments where hundreds of heterogeneous DB instances coexist.

- Statistical Sampling-based Ultra-lightweight Non-intrusive Profiling
  - Overcoming the limitations of existing full scan methods (service load generation) for large-scale DBs by applying data distribution-based sampling techniques to determine PII presence with minimal queries.
- High-Precision Semantic Classification via Metadata-Data Context Cross-Analysis
  - Accurately identifying personal information even in unclear columns (e.g., 'col_01', 'temp_data') by cross-analyzing the correlation between 'Column Name (Metadata)' and 'Actual Data Value' using sLLM, rather than simply looking at column names or data patterns (Regular Expressions).
- Real-time Tracking of Multi-Instance Schema Changes & Dynamic Asset Management
  - Centrally detecting table creation/modification (Schema Drift) events occurring frequently across numerous DB instances and immediately performing PII inspection on newly created assets to fundamentally block the generation of 'Shadow Data'.

3. Development of High-Speed Threat Detection Technology Native to On-Premise Appliances based on sLLM Knowledge Distillation

A technology to distill high-quality threat analysis knowledge generated by sLLM on high-performance GPU servers (Offline) into a lightweight model, enabling real-time precise detection of obfuscated threats on CPU-based security appliances (Online).

- High-Density Threat Data Generation & Automated Labeling using Offline sLLM (Teacher)
  - Mass production of high-density training data where sLLM interprets obfuscated attack codes and includes 'Execution Intent' and 'Threat Score (Soft Label)'. (Creating textbooks to teach the Student model, performed in an offline environment (Lab) that does not consume appliance resources.)
- Encoder-based Student Model Training Optimized for Appliance Environments
  - Adopting an Encoder (TinyBERT, etc.) structure favorable for CPU processing while inheriting the Teacher's knowledge.
  - Training the Student model to mimic the Teacher's reasoning ability by defining the Teacher model's output (probability distribution) as a Loss Function.
  - Through this, the model size is reduced to 1/50 while maintaining detection accuracy.
- Inference Optimization & Acceleration Technology for Maximizing CPU-based Appliance Performance
  - Technology to accelerate the trained Student model using various methods such as Structural Pruning, Adaptive Inference Control, and Hardware-Native Acceleration to ensure sufficient speed on CPU-based appliances.

R&D Target Specifications

1. Autonomous Threat Hunting & Response

Metrics	Target	Remark / Measurement Method
Exploit Generation Rate	> 80%	Success rate of generating valid test attack codes when analyzing new CVEs (CVSS 9.0+)
Zero-day Readiness	< 30 mins	Time taken from CVE disclosure to attack simulation and detection rule deployment
Timeline Accuracy	> 95%	Consistency of attack stages and times reconstructed by AI compared to actual attack scenarios (Red Teaming)
Mean Time To Respond (MTTR)	< 1 min	Time taken for AI to complete blocking/isolation measures after threat detection (Reduced from 30 mins to 1 min)

2. Massive PII Discovery & Observability

Metrics	Target	Remark / Measurement Method
PII Classification Precision	> 95%	Accuracy of identifying PII within obfuscated column names and unstructured data (Based on F1-Score)
Scanning Overhead	< 5%	Additional CPU/Memory usage of the target DB during PII profiling (Non-intrusive)
Drift Detection Latency	Within 10 mins	Time to detect new table/column creation and complete PII determination
Scanning Throughput	1,000 cols/min	Number of columns automatically classified per minute (When applying Metadata + Sampling method)

3. High-Speed On-Premise Detection

Metrics	Target	Remark / Measurement Method
Inference Throughput	> 10,000 TPS	Number of web logs/transactions processable per second in x86 CPU (32 Core) environment
Accuracy Retention	> 95%	Performance retention ratio of the Student model compared to the detection accuracy of the Teacher model (sLLM)
Detection Latency	< 10ms	Time taken for AI to determine maliciousness upon single packet/log ingress

R&D Plan

- Phase 1: Feasibility Study & Joint Agreement (Months 1-6)

  This phase aims to establish a robust collaborative infrastructure through securing technical feasibility via IP-R&D and detailed design, thereby fully preparing for the Gate Review.

  - IP-R&D & Detailed Design (Strategic Design)
    - IP-R&D Strategy Formulation: Establishing strategies to analyze global patent infringement risks and derive avoidance designs for core technologies—specifically 'Autonomous Rule Generation' and 'Self-Validation'—in linkage with TIPA/MSS support programs.
    - Technical Interface Definition: Deriving the final technical agreement regarding API

specifications and performance goals (KPIs) between the Attack Generation Module and the Detection Engine.

- Phase 2: Main R&D & Implementation (Months 7-36)

  During the main research period, we will conduct substantial joint research through personnel exchanges and regular technical seminars beyond simple video conferencing.

  - [Year 1: M7 ~ M18] Core Logic Development & Technical Exchange
    - Key Research:
      - CVE-based Attack Generation & Self-Validation: Development of an Autonomous Agent (Alpha) where sLLM generates attack codes and simulates/validates them in a virtual environment.
    - International Collaboration:
      - Suitability Tests: Perform suitability tests of early models in the testbed environment.
  - [Year 2: M19 ~ M30] On-Premise Optimization & Integration
    - Key Research:
      - sLLM Knowledge Distillation & CPU Acceleration: Transferring Teacher model knowledge to a lightweight Student model and integrating AVX-512 based quantized inference engine.
    - International Collaboration:
      - Joint Intensive Development Workshop: Researchers gather in one location during the core development period to perform intensive tuning to achieve 'Model Compression Rate' and 'Inference Speed' goals.
  - [Year 3: M31 ~ M36] Validation
    - Key Research:
      - Field Test: Validation in large-scale traffic environments.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1. Implementation of Active Threat Hunting & Self-Validating 'Agent' (Core Agent Development) ● [Attack Generation Agent] Development of an intelligent agent software that operates on the developing framework, analyzing CVE information, Campaign Report to generate and execute attack codes. ● [Agentic Workflow Design] Design of ReAct-based reasoning logic and prompts to enable the agent to perform self-validation through the process of 'Plan → Act → Observe', going beyond simple API calls. 2. sLLM Knowledge Distillation & Hardware Acceleration Optimization Experiments ● [Distillation Pipeline] Construction and experimentation of a pipeline to transfer the inference knowledge of the Teacher model (sLLM) to a lightweight Student model. ● [HW Acceleration Experiment] Experimenting with accuracy loss and speed improvements when applying Quantization in CPU (AVX-512) environments, and deriving optimal lightweighting parameters. 3. Generation of High-Quality Synthetic Data & Scientific Performance Verification ● [Synthetic Data Generation] Using the Teacher model to generate high-quality synthetic training datasets containing rare threat patterns or obfuscated attack examples (IP belongs to the Lead Agency). ● [Scientific Validation] Defining quantitative performance metrics for detection rates and generation success rates of the developed agent/model, and verifying experimental results.
Expected R&R of Korean SME Partner	1. Provision of High-Performance Big Data Platform & Agent Framework ● [Foundation] Providing a high-performance unified security platform based on Sonar/Forensic where the AI Agent will be deployed. ● [Agent Framework] Developing and providing the API and execution Infrastructure that allows the Agent to simulate attacks (Sandbox) and deploy/test detection rules. 2. Integration with Commercial Engine & System Implementation ● [Porting] Porting the optimized models and agent logic verified by the research institute into the commercial engine (C++/Java) for productization. ● [Integration] Developing functional integration so that the Agent's threat judgment results trigger automated response workflows in the existing SOAR system. 3. Large-scale Validation ● [Agent Validation] Verifying the validity of rules generated by the Agent and the accuracy of autonomous responses in real-world operational environments.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The 'On-Premise Autonomous Security Agent' technology secured through this international joint research will revolutionize the existing security operations paradigm. It will create substantial economic and social value by solving labor shortages and reducing costs, alongside technological advancements. 1. Innovation in Security Operations Paradigm: "Passive Detection" → "Active Autonomous Hunting" ● [Status Quo] Existing security systems remain at a passive level of analyzing logs after an attack occurs, having fundamental limitations in responding to unknown Zero-day threats. ● [Innovation] The 'Self-Adversarial Validation' technology of this project internalizes the concept of 'Offensive Security', where AI creates attack scenarios and tests defense barriers from a hacker's perspective. This signifies a technological leap to an Active Autonomous Security System that proactively identifies and blocks threats instead of waiting for them. 2. Removing Barriers to Entry & Universalization of On-Premise AI ● [Status Quo] Adopting high-performance AI like sLLM requires expensive GPU server infrastructure, creating a 'Technology Gap' where SMEs with limited budgets or public institutions using closed networks cannot implement such technologies. ● [Innovation] Through Knowledge Distillation and CPU Acceleration (AVX-512) optimization technologies, sLLM-level intelligence can run on existing general x86 servers (Appliances) without separate GPU equipment. This removes the biggest obstacle—hardware constraints—providing a universal environment where all organizations can utilize advanced AI security technology without cost burdens. 3. Dramatic Reduction of TCO by Resolving Talent Shortage ● [Socio-Economic Problem] The global shortage of cybersecurity experts causes skyrocketing labor costs, and companies suffer from security gaps due to the inability to hire personnel. This directly leads to increased Operating Expenses (OPEX) and risk management costs. ● [Solution & Effect] The developed Autonomous Security Agent automates over 90% of tasks performed by Junior Analysts (ticket analysis, rule generation, basic response), creating the effect of hiring 'Digital Experts'. · Workforce Efficiency: Allowing a few experts to focus only on high-value threat analysis, maximizing the processing capacity of the SOC without additional hiring. · Cost Reduction: By eliminating not only labor costs but also the CAPEX for building expensive GPU infrastructure, it creates an economic effect of reducing the customer's Total Cost of Ownership (TCO) for security operations by more than 50%.

6. References

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